

MTT - I

Academic Year 2024-25 (Odd Semester)

SET-A

Course	:	B.Tech.	Date	:	7/11/2024
Semester/ Section	:	I/ A,B,C,D,E,F,G,H,I,J & K	Time Duration	:	1:30 Hrs.
Subject & Subject Code	:	Engineering Physics /1FY2-02	Max. Marks	:	50

Course Outcomes	
CO1	Students will be able to explain the basic concepts, theoretical principles and practical applications of interference, diffraction phenomena and their related optical devices in visible range and X-ray diffraction by crystals (i.e., Bragg's law).
CO2	Students will be able to acquire knowledge of fundamental concepts, principles of quantum mechanics to understand numerous atomic and molecular scale phenomena.

Q. No.	CO	Questions	Marks
PART- A: Attempt All Questions (10x2 = 20 Marks)			
1.	1	Define Interference of light.	2
2.	1	Why the Newton's rings are circular in shape?	2
3.	1	Explain Rayleigh's criterion of just resolution of two spectral lines.	2
4.	1	Define Diffraction of light.	2
5.	1	What is the function of compensatory plate?	2
6.	2	Define wave function.	2
7.	2	Write Hamiltonian operator in three dimensions.	2
8.	2	Define degeneracy of energy level.	2
9.	2	What is zero point energy state?	2
10.	2	Write down the orthogonal and normalization conditions.	2
PART-B: Descriptive /Analytical / Problem Solving/ Design questions Attempt Any Three Questions (3x5=15 Marks)			
1.	1	Define construction of Newton's rings and derive an expression for Diameter of dark rings.	5
2.	1	Derive an expression for the intensity of diffracted light in the Fraunhofer's diffraction	5

		due to single slit	
3.	1	Evaluate the distance between two successive positions of a movable mirror of a Michelson interferometer giving distinct fringes in case of sodium having lines of wavelengths $5600 \text{ \AA}$ and $5610 \text{ \AA}$.	5
4.	1	The distance between the first and the sixth minima in the diffraction pattern of a single slit is 0.5 mm . The screen is 0.5 m away from the slit. If the wavelength of light used is $5000 \text{ \AA}$, evaluate the slit width.	5

PART-C: Descriptive /Analytical / Problem Solving/ Design questions

Attempt Any Three Questions (3x5=15 Marks)

1.	2	Derive time independent Schrodinger's equation.	5
2.	2	Find the probability that a particle in a box 'a' width can be found between $x=0$ and $x=a/n$, when it is n^{th} state.	5
3.	2	In a long chain molecule of length $5 \text{ \AA}$, electron may be treated as to be free to move along the length. Calculate zero point energy, the energy gap between first two energy states of the electron and also the wavelength of absorption line arising from this length.	5
4.	2	Prove that the wave functions are orthogonal to each other if particle is in 1-D box.	5